

Exercise (Last slide)

```
ecost<- c(96, 171, 202, 178, 147, 102, 153, 197, 127, 82, 157, 185, 90, 116, 172, 111, 148, 213, 130, 165, 141, 149, 206, 175, 123, 128, 144, 168, 109, 167, 95, 163, 150, 154, 130, 143, 187, 166, 139, 149, 108, 119, 183, 151, 114, 135, 191, 137, 129, 158)
```

```
length(ecost)
```

```
[1] 50
```

```
library(psych)
```

```
describe(ecost)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X1	1	50	147.06	31.69	148.5	146.98	31.13	82	213	131	0.01	-0.7	4.48

```
cv<- sd(ecost)/mean(ecost)*100
```

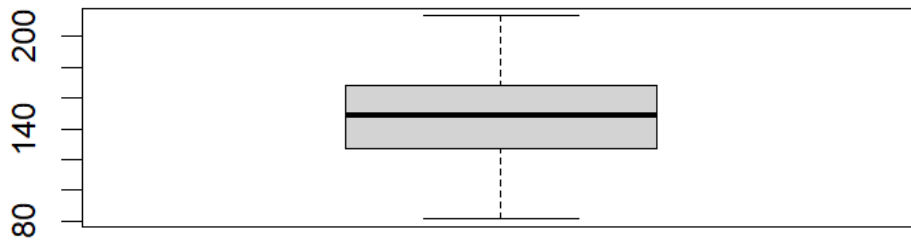
```
cv
```

```
[1] 21.54996
```

```
summary(ecost)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
82.0	127.2	148.5	147.1	167.8	213.0

```
boxplot(ecost)
```



```
##Outlier Checking
```

```
quant_ec <- quantile(ecost)
```

```
quant_ec
```

0%	25%	50%	75%	100%
82.00	127.25	148.50	167.75	213.00

```
lowerfence<- quant_ec[2]-1.5*IQR(ecost)
```

```
lowerfence
```

```
25%
```

```
66.5
```

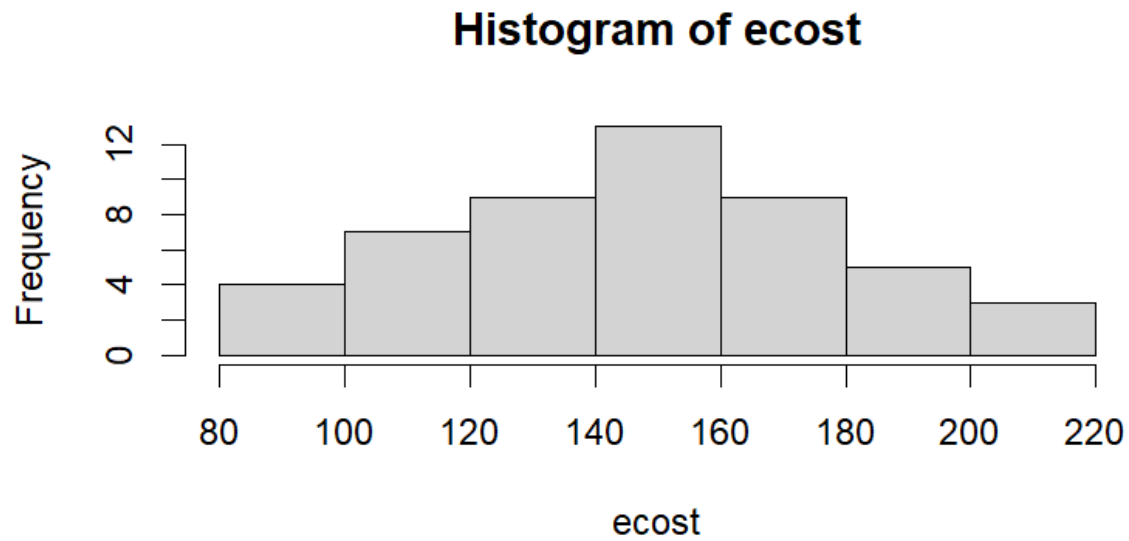
```
upperfence<- quant_ec[4]+1.5*IQR(ecost)
```

```
upperfence
```

```
75%
```

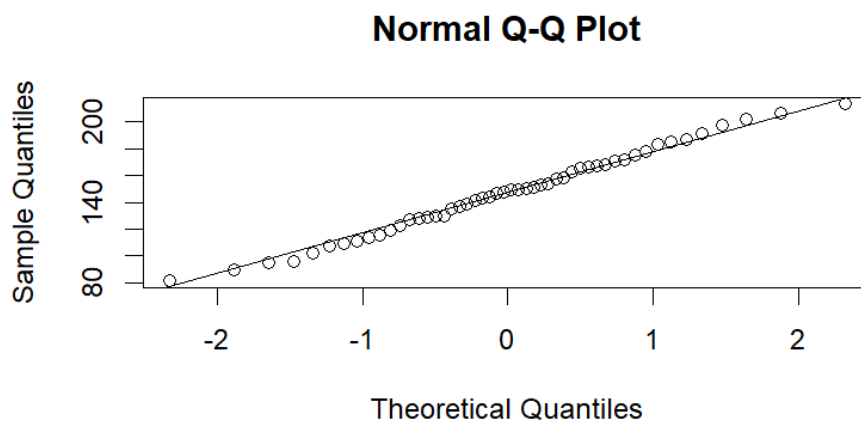
```
228.5
```

```
hist(ecost)
```



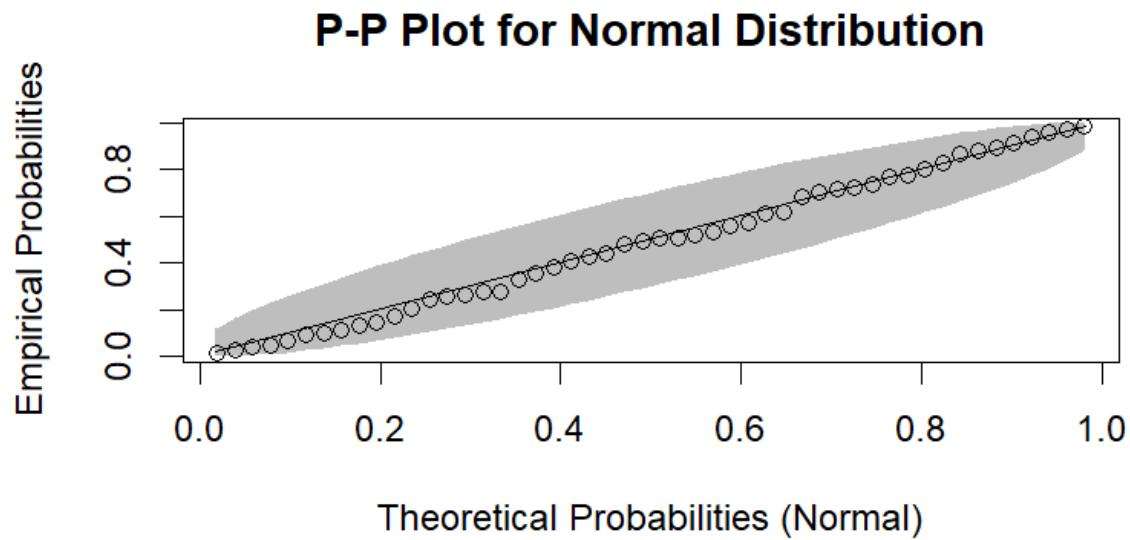
```
qqnorm(ecost)
```

```
qqline(ecost)
```



```
library(qqconf)

pp_conf_plot(ecost, pdist = pnorm, main = "P-P Plot for Normal Distribution",
             xlab = "Theoretical Probabilities (Normal)",
             ylab = "Empirical Probabilities")
```



```
shapiro.test(ecost)
```

Shapiro-Wilk normality test

data: ecost

W = 0.98977, p-value = 0.9408

Z Table

```
pnorm(-1.645)
```

```
[1] 0.04998491
```

Skewness: Skewness measures the symmetry of a variable's distribution. If the distribution stretches toward the right or left tail, it's skewed. Negative skewness indicates more larger values, while positive skewness indicates more smaller values. A skewness value between -1 and +1 is excellent, while -2 to +2 is generally acceptable. Values beyond -2 and +2 suggest substantial nonnormality (Hair et al., 2022, p. 66).

Kurtosis: Kurtosis indicates whether the distribution is too peaked or flat compared to a normal distribution. Positive kurtosis means a more peaked distribution, while negative kurtosis means a flatter one. A kurtosis greater than +2 suggests a too peaked distribution, while less than -2 indicates a too flat one. When skewness and kurtosis are close to zero, it's considered a normal distribution (Hair et al., 2022, p. 66). In the rare scenario where both skewness and kurtosis are zero, the pattern of responses is considered a normal distribution.

References 3 • Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2022). A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM) (3 ed.). Thousand Oaks, CA: Sage.